Customer Churn Prediction for a Telecom Company

PREDICTING WHICH CUSTOMERS ARE LIKELY TO LEAVE USING MACHINE LEARNING

PRESENTED BY: MOHSIN KHAN | DATA SCIENCE INTERN

Why Predicting Churn Matters?

- • Losing customers = losing revenue 💸
- • Early prediction = targeted retention strategies 📲
- • Goal: Predict churn and identify key drivers using ML

Understanding the Data

- • ~2000 customer records from a telecom company
- • Features include:
- - Customer demographics
- - Subscription info
- - Payment details
- - Churn label (target)

Exploratory Data Analysis (EDA)

- • Detected class imbalance → Churn = 26%
- • Higher churn observed in:
- - Month-to-month contracts
- - Customers with fiber optic internet
- - High monthly charges

Data Cleaning & Preprocessing

- • Handled missing values
- • Encoded categorical variables (Label + OneHot)
- • Applied SMOTE for balancing classes
- • Feature scaled where needed

Feature Engineering

- • Created tenure in years
- • Combined MonthlyCharges + tenure → avg lifetime value
- • Selected relevant features for training

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.63	0.40	0.30	0.34	0.56
Random Forest	0.62	0.30	0.13	0.18	0.57
Gradient Boosting	0.62	0.37	0.27	0.32	0.55
Tuned RF (final)	0.63	0.38	0.23	0.28	0.57

Tuned Random Forest Results

- • Accuracy: 63%
- • ROC AUC: 0.57
- • Confusion Matrix:
-   - True Negatives: 221
-   - True Positives: 29

Insights & Business Impact

- • Customers with month-to-month contracts → high churn risk
- • Higher charges = higher churn probability
- • Tenure is a strong negative predictor of churn
- Actionable Suggestion:
- “Focus retention offers on high-churn segments like new customers and fiber users with month-to-month contracts.”

Next Steps + Thank You!

- • Improve recall using cost-sensitive learning or XGBoost
- • Consider customer sentiment data in future modeling
- • Build dashboard for churn monitoring
- 🧠 Questions?
- 🧑‍💻 Let's talk data!